

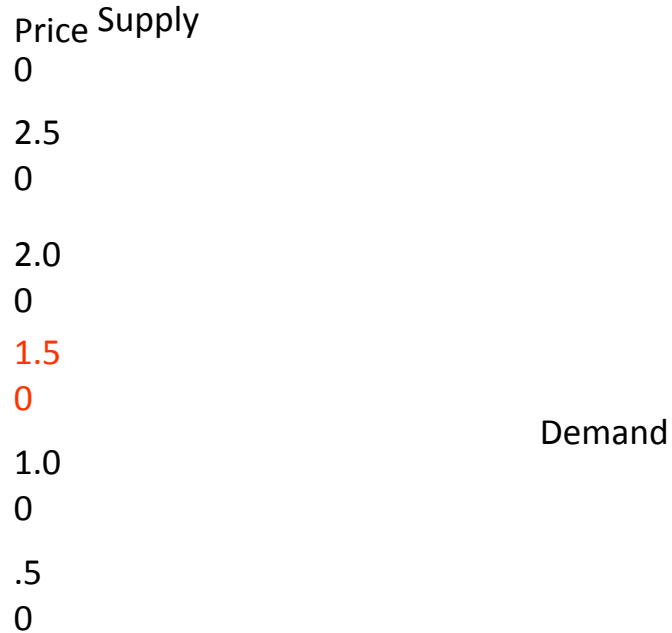
Market Equilibrium

Market Equilibrium

- *Equilibrium* in a market occurs when the price balances the plans of buyers and sellers. ●
- *Equilibrium price* is the price at which the quantity demanded equals the quantity supplied. ●
- *Equilibrium quantity* is the quantity bought and sold at the equilibrium price.

Market

Equilibrium 3.0



Quantity

0 5 10 15 20

Market Equilibrium

- A market moves towards its equilibrium because:
 - Price regulates buying and selling plans -
Price adjusts when plans do not match
- When price is too low: $Q_d > Q_s$ - shortage

- When price is too high: $Q_s > Q_d$ - surplus
- Price adjusts so that $Q_d = Q_s$ - equilibrium

Price adjustments

When price is too low: $Q_d > Q_s$ - shortage

- Producers raises the price to meet the demand of the unsatisfied consumers
- As price is increased, producers increases supply -
The rising price reduces the shortage and reaches the equilibrium because it decreases the quantity

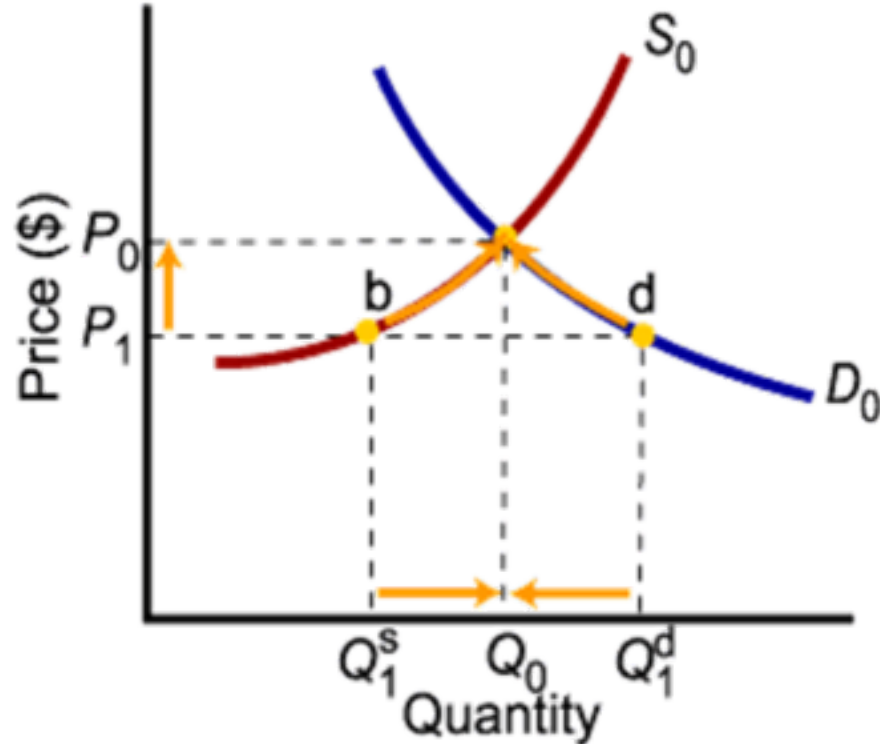
demand and increases the quantity supplied

Shortages

- If the price is too low, the quantity demanded exceeds the quantity supplied.

People are willing to pay more for the good.

- To eliminate this



shortage,
sellers will raise the price,
increasing the quantity
supplied and reducing the
quantity demanded.

Price adjustments

When price is too high: $Q_s > Q_d$ - surplus

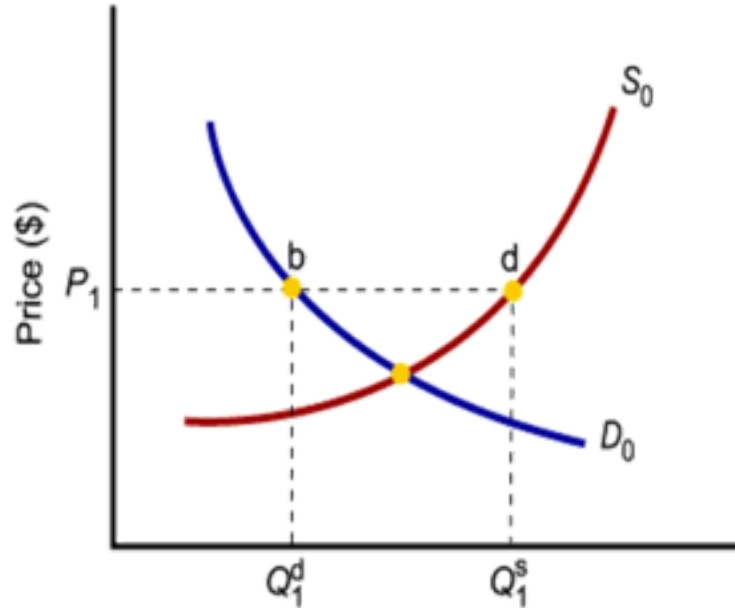
- Producers lowers the price as they are unable to sell the quantities of the good they planned to sell - As price is decreased, producers decreases supply - The

falling price reduces the surplus and reaches the equilibrium because it increases the quantity demanded and decreases the quantity supplied

Surpluses

- If the price is too high, the quantity supplied exceeds the quantity demanded. Inventories pile up.

- To eliminate this

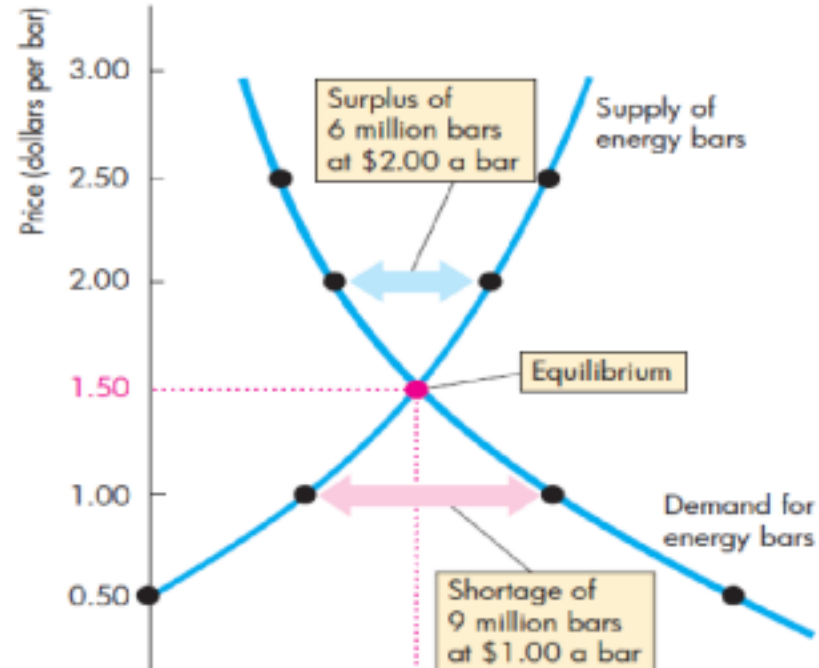


surplus,

sellers will lower the price,
reducing quantity supplied
and increasing quantity
demanded.

- A Shortage Forces the Price Up ○ If demand exceeds supply, sellers will raise price, decreasing quantity demanded.
- A Surplus Forces the Price Down ○ If supply exceeds demand, sellers will see their inventories of unsold

FIGURE 3.7 Equilibrium



goods piling up and will cut price to sell them.

Demand Increases if:

- The price of a substitute rises
- The price of a complement falls
- The expected future Price or income rises
- Income rises*
- No of consumers/ population increases

*considering normal goods

Demand Decreases if:

- The price of a substitute falls ■
- The price of a complement rises
- The expected future Price or income falls
- Income falls*
- No of consumers/ population decreases

*considering normal goods

Changes in demand

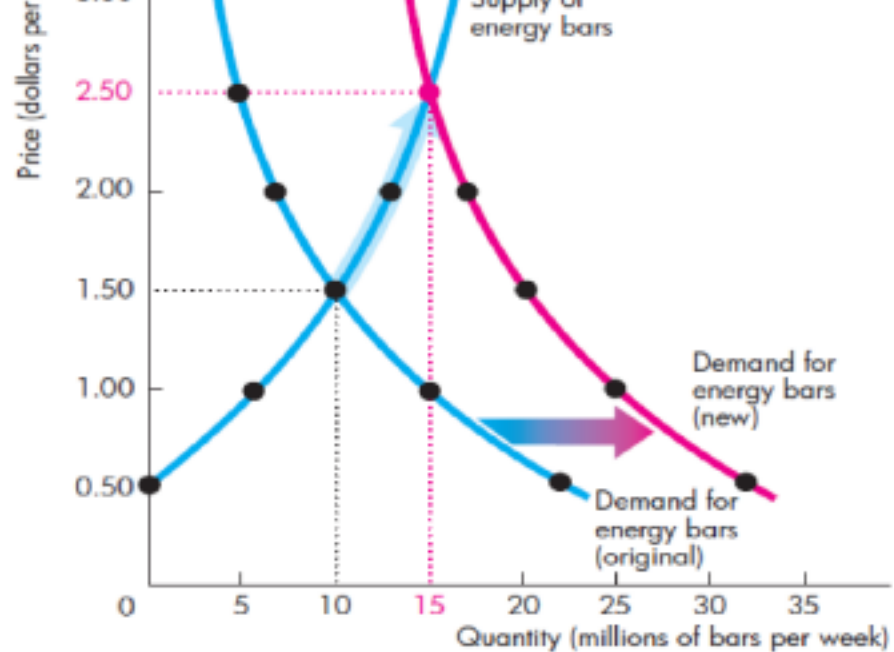
- When demand increases, both the price and the quantity increase.
- When demand decreases, both the price and the quantity decreases.

Effect of an Increase in Demand on Equilibrium Price and Quantity

- An increase in demand shifts the demand curve to the right (outward shift in demand).

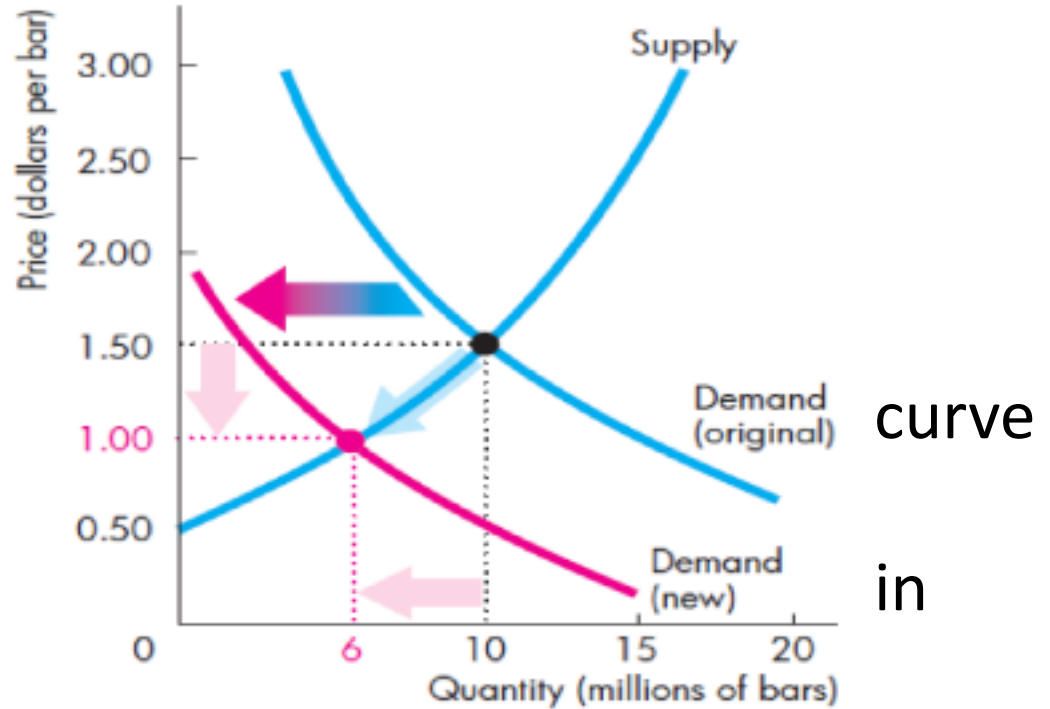
- The new equilibrium price and quantity are higher.

Effect of a decrease in Demand on



Equilibrium Price a

- A decrease in demand shifts the demand curve to the left (inward shift demand).



- The new equilibrium price and quantity are lower.

Supply Increases if:

- The price of a factor of production (input price, technology) falls
- Favorable govt. policy
- The expected future price falls
- The number of suppliers increases
- The price of a substitute in production falls
- The price of a complement in production rises
- A favorable natural event increases production

- The price of a factor of production (input price, technology) rises
- Unfavorable govt. policy
- The expected future price falls
- The number of suppliers of increases
- The price of a substitute in production rises
- The price of a complement in production falls
- A unfavorable natural event decreases production

Supply Decreases if:

Changes in supply

- When supply increases, the quantity increases and the price falls.
- When supply decreases, the quantity decreases and the price rises.

Effect of An Increase in Supply on Equilibrium Price

- An increase in supply shifts the supply

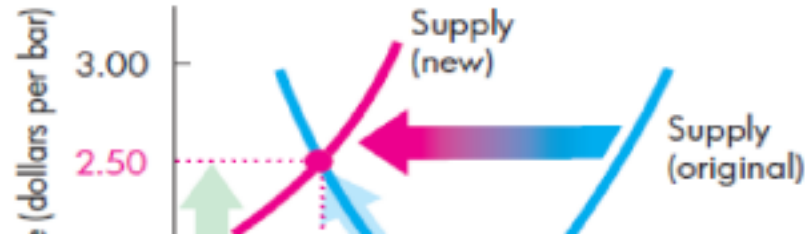


curve to
the right (outward shift in
supply).

- The new equilibrium price is lower, but the equilibrium quantity is higher.

Effect of An Decrease in Supply on Equilibrium Price

- A decrease in supply shifts



the supply curve to the left
(inward shift in supply).

- The new equilibrium price is higher, but the equilibrium quantity is lower.

A Change in Both Demand and Supply

- There are four possible scenarios
 1. Both demand and supply increase
 2. Both demand and supply decrease
 3. Demand increases and supply decreases

4. Demand decreases and supply increases

- Effects-

- Both curves shift.
- Price and quantity change will depend on how each curve shifts on which direction.

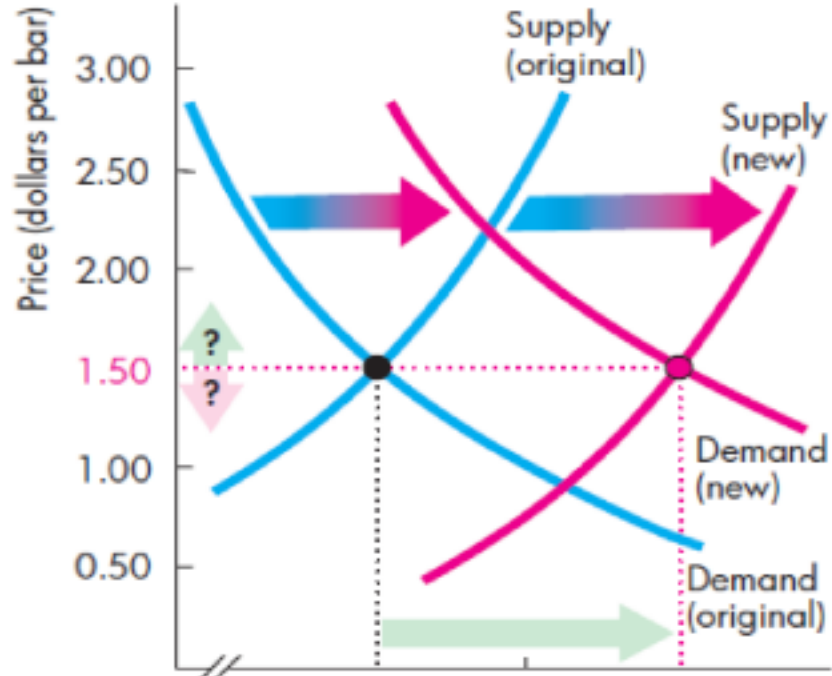
Changes in demand and supply

- In same direction,
 - When both demand and supply increase, the quantity increases and the price might increase, decrease or remain the same.

- When both demand and supply decrease, the quantity decreases and the price might increase, decrease or remain the same.

Both demand and supply

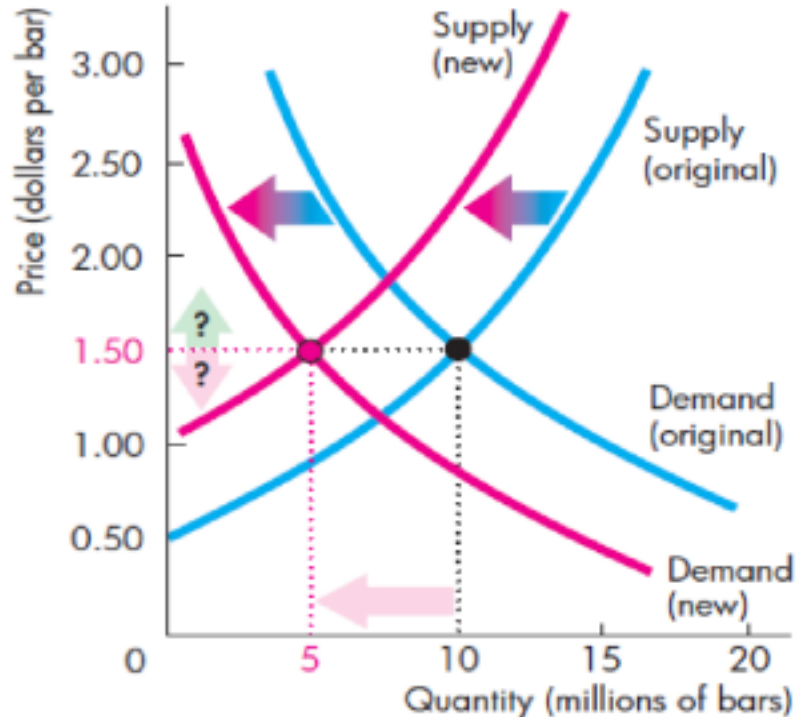
- **If demand and supply increase**, both the demand and supply curves shift to the right (outward).
- **Effect on equilibrium price and quantity**
 - The new equilibrium quantity will be higher.



- The new equilibrium price may be higher, lower, or it may remain the same.

Both demand and supply

- If demand and supply decrease, both the demand and supply curves shift to the left (inward).
- Effect on equilibrium price and quantity
 - The new equilibrium quantity will be lower.
 - The new equilibrium price may be higher, lower, or it may remain the



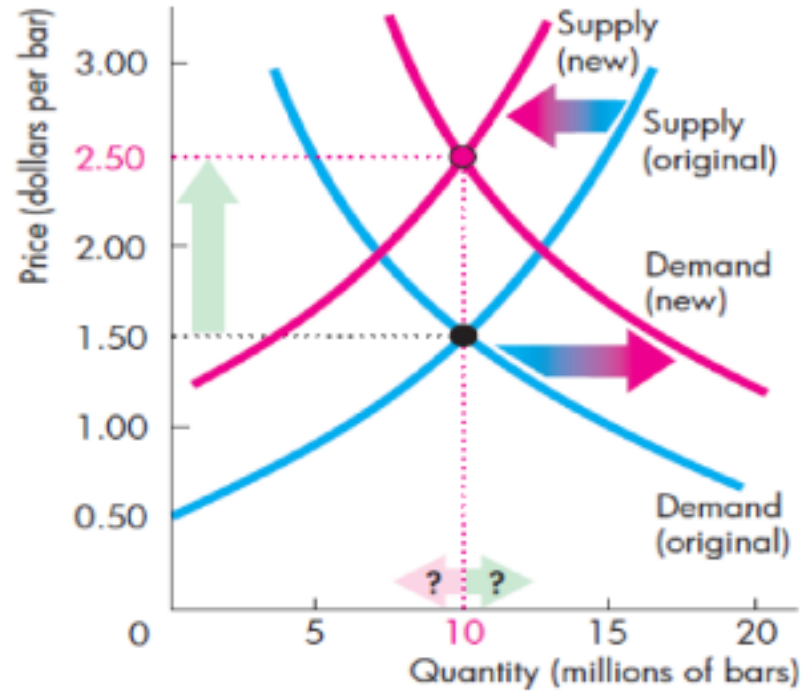
same.

Changes in demand and supply

- In opposite directions,
 - When demand increases and supply decreases, the price rises and the quantity might increase, decrease or remain the same.
 - When demand decreases and supply increases, the price falls and the quantity might increase, decrease or remain the same.

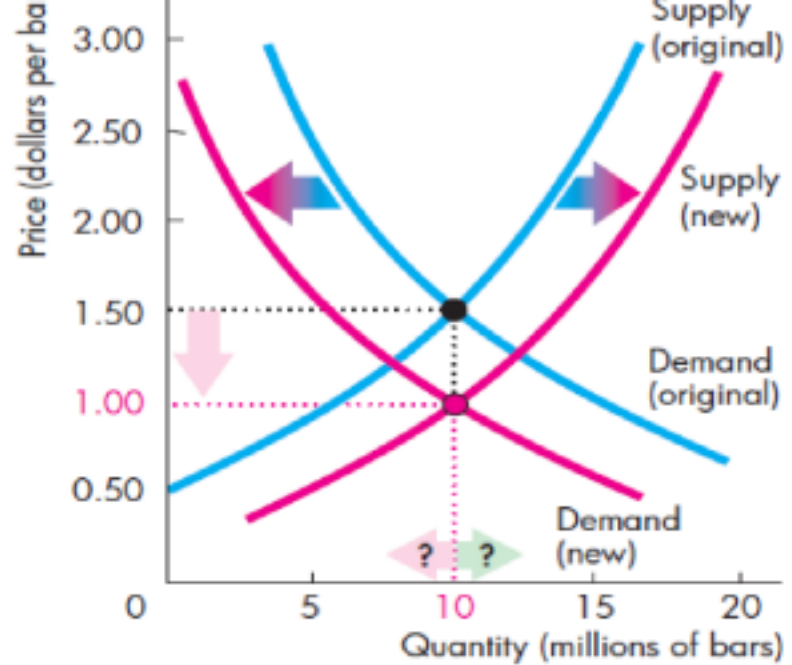
Demand increases and supply decreases

- **When demand increases**, demand curve shifts to the right (outward).
- **When supply decreases**, supply curves shift to the left (inward).
- Effect on equilibrium price and quantity
 - The new equilibrium price will be higher.
 - The new equilibrium quantity may be higher, lower, or it may remain the same.



Demand decreases and supply increases

- **When demand decreases**, demand curve shifts to the left (inward).
- **When supply increases**, supply curves shift to the right (outward).
- **Effect on equilibrium price and quantity**
 - The new equilibrium price will be lower.
 - The new equilibrium quantity may be higher, lower, or it may remain the same.



Example: Market for Bottled Water in the US

1. A new study says that tap water is unsafe.
2. A new zero-calorie sports drink is invented. The new drink is a substitute for bottled water.
3. European water bottlers buy springs and open plants in the United States.
4. Drought dries up some springs in the United States.

Scenario 01

Effects of Changes in Demand

Scenario 1: A new study says that tap water is unsafe.

In the market for bottled water:

- The demand for bottled water increases, the demand curve shifts rightward.
- The equilibrium price rises.
- The quantity supplied increases along the supply curve. -
Equilibrium quantity increases.

Scenario 02

Effects of Changes in Demand

Scenario 2: A new zero-calorie sports drink is invented. The new drink is a substitute for bottled water.

In the market for bottled water:

- The demand for bottled water decreases, the demand curve shifts leftward.
- The equilibrium price falls.
- The quantity supplied decreases along the supply curve. - Equilibrium quantity decreases.

Scenario 03

Effects of Changes in Supply

Scenario 3: European water bottlers buy springs and open plants in the United States.

In the market for bottled water:

- The supply for bottled water increases because of more suppliers, the

supply curve shifts rightward.

- The equilibrium price falls.
 - The quantity demanded increases along the demand curve. -
- Equilibrium quantity increases.

Scenario 04

Effects of Changes in Supply

Scenario 4: Drought dries up some springs in the United States.

In the market for bottled water:

- The supply for bottled water decreases, the supply curve shifts leftward. -
- The equilibrium price rises.
- The quantity demanded decreases along the demand curve. -

Equilibrium quantity decreases.

Reminders

When only demand changes:

- The supply curve does not shift.
- But there is a change in the quantity supplied (because price changes).

When supply changes:

- The demand curve does not shift.
- But there is a change in the quantity demanded.

Practice Example: Equilibrium price and quantity

Suppose the demand and supply functions for ice cream are given by: **$P = 400 - 2Q_d$** **$P = 100 + Q_s$**

1. Find the equilibrium price and quantity.
2. If government regulates the market for ice-cream and fixes price at TK 150, will there be a shortage or surplus in the market?
3. Calculate the amount of surplus/shortage.

Relative magnitude of change question for practice

- There is a long holiday coming during 21st February. Holiday goers are seeing this opportunity and planning to buy tickets before the prices go up. Due to recent wage/salary cuts of pilots, most of the pilots have gone on a strike. Say that the effect of demand is more than the effect of supply.
- Draw the demand and supply diagram for the market of air tickets and describe change in market equilibrium.

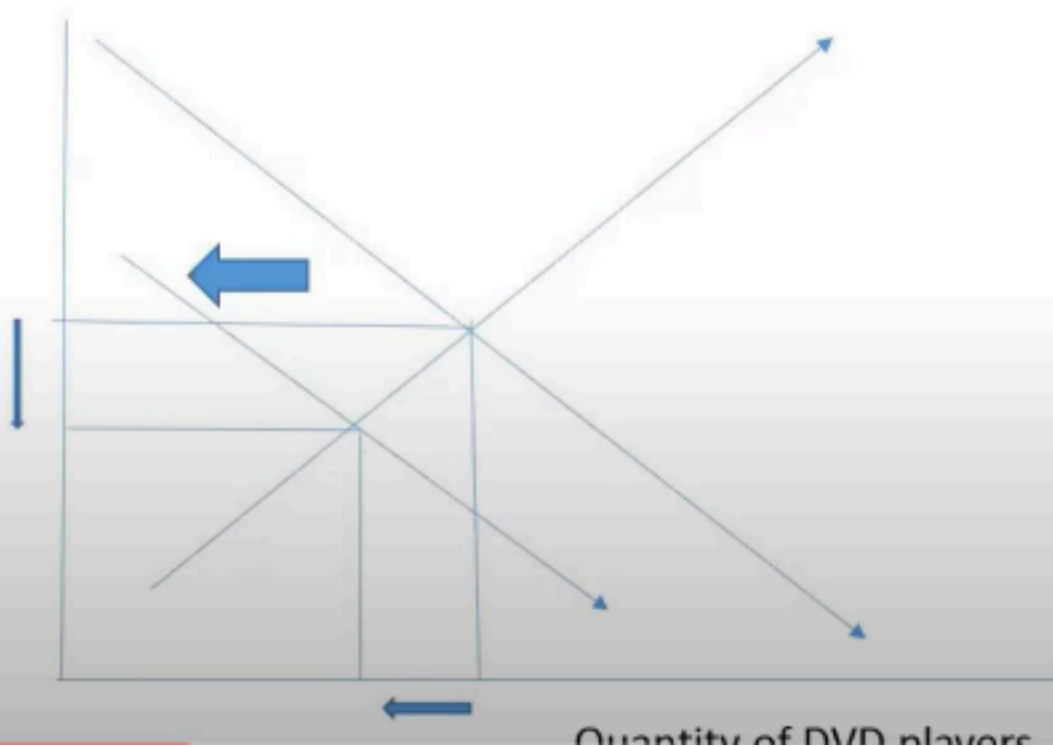
Predicting effect on price and quantity

What is the effect on the price of a DVD player and the quantity of DVD players sold if:

- a** The price of a DVD rises?
- b** The price of a DVD falls?
- c** The supply of DVD players increases?
- d** Consumers' incomes decrease?
- e** The wage rate of workers who produce DVD players increases?
- f** The wage rate of workers who produce DVD players rises and at the same time the price of a DVD falls. The effect of increasing wage rate is higher than the effect DVD price falling.

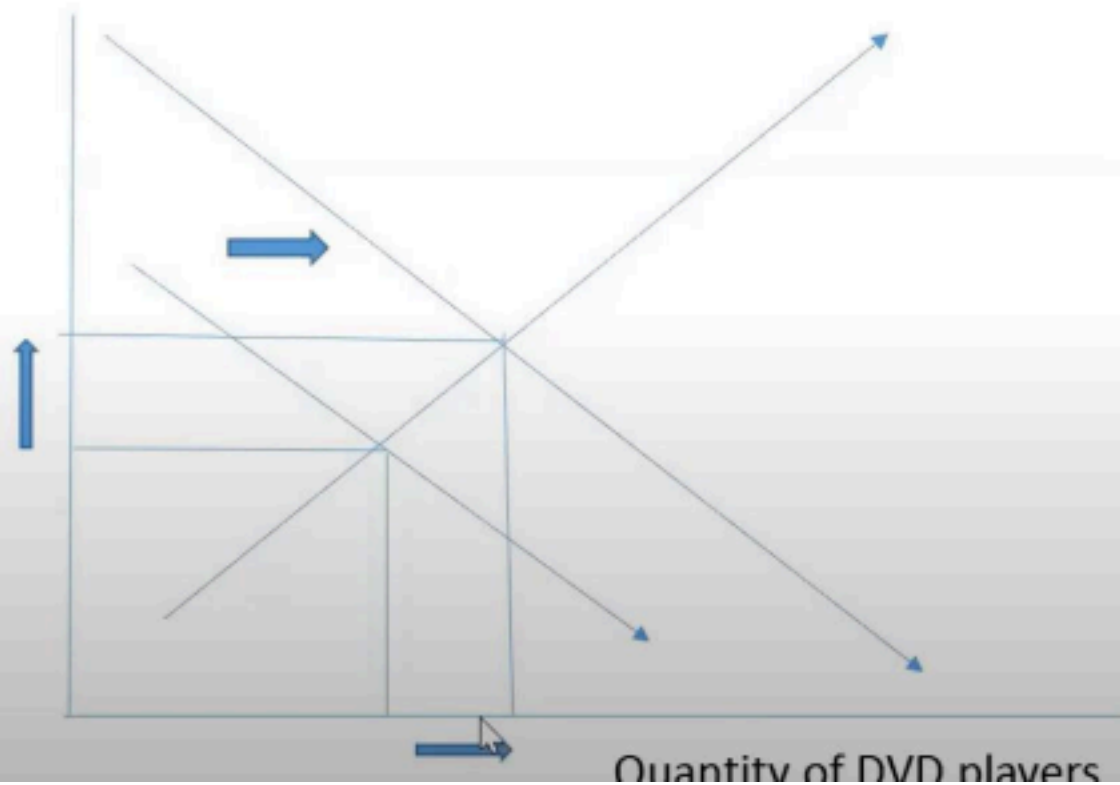
- a. The price of a DVD rises?

Price of DVD
players



- b. The price of a DVD falls?

Price of DVD
players



Quantity of DVD players









Summary

- *Key concepts:*
 - Market equilibrium
 - Price adjustments
 - Changes in demand and supply
 - Graphical and mathematical interpretations
- *Reading: Chapter 3 of “Economics” by Parkin, Powell, Mathews 6th Ed.*

